Cross-Patient Speech Decoding from Intra-Operative µECoG

Technical Memo: Research Plan

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March 17, 2026

Synthesizes 18 papers on cross-patient neural speech decoding into a two-phase implementation plan. Phase 2 modernizes Spalding's PCA+CCA+SVM to the field-standard end-to-end pipeline. Phase 3 introduces two architectural innovations—each absent from the neural speech BCI field—while keeping training field-standard.

Supporting files: `implementation_plan.md` · `research_synthesis.md` · `future_directions.md`

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1 Summary

I have reviewed 18 papers on cross-patient neural speech decoding and designed a two-phase plan to replace the PCA+CCA+SVM pipeline (0.31 bal acc) with the modern field-standard architecture, then add two novel architectural innovations. The plan is conservative: 8 of 10 pipeline components are field-standard with multi-paper support. The 2 novel components are independently ablated.

Bottom line: Phase 2 (field standard) is a guaranteed improvement. Phase 3 adds two architectural ideas absent from the field—per-patient spatial conv and an articulatory decomposition CTC head. Training is identical in both phases (standard multi-patient SGD, same as Singh 2025).

1.1 Papers Reviewed

Spalding 2025, Duraivel 2023/2025, Singh 2025, Boccato 2026, Levin 2026, BIT (Zhang 2026), wav2vec ECoG 2024, Dual Pathway 2025, Chen 2024, Chen 2025 (SwinTW), MIBRAIN 2025, Wu 2025, Willett 2023, Metzger 2023, Littlejohn 2025, Nason 2026, Qian 2025.

2 What the Literature Tells Us

2.1 Architecture Consensus

Every successful cross-patient speech BCI from the last two years uses:

Per-patient input layer ⟶ Shared recurrent backbone ⟶ CTC/sequence loss

Used by Willett 2023, Metzger 2023, Boccato 2026, Levin 2026, Nason 2026, Singh 2025 (all GRU/LSTM + CTC). Littlejohn 2025 evolves this to RNN-T for streaming. BIT 2026 uses a transformer backbone + CTC but the same per-patient read-in pattern. The per-patient layer absorbs cross-patient variation; the shared backbone learns patient-invariant temporal dynamics; CTC handles variable-length alignment.

2.2 Key Findings

Singh 2025 — closest precedent. 25 sEEG patients, per-patient Conv1D, shared BiLSTM 2×64, CTC. Group PER 0.49 vs single 0.57 (~14% relative improvement). Only 2 layers of hidden size 64 for 36 phonemes (of 44 English)—our 9-phoneme task is far simpler.

Boccato 2026 — learned affine transforms are predominantly diagonal (per-channel gain). **Caveat:** this is Utah-specific (fixed stereotactic placement, same orientation). uECOG has variable surgical placement/orientation → cross-patient variation includes *spatial remapping*, not just gain. This strengthens the case for spatial conv over linear read-in.

Levin 2026 — per-patient input layers are essential (shared input makes transfer *worse*). 30% source replay during fine-tuning prevents catastrophic forgetting.

BIT (Zhang 2026) — SSL ≈ supervised for same-subject (Table 9). SSL advantage is cross-subject only. Supervised cross-patient with per-patient layers (Singh) works at our scale.

Metzger 2023 / Littlejohn 2025 (Chang lab) — Conv1d → GRU on 253-ch ECoG. Metzger: 4-layer BiGRU H=500, CTC, 50 Hz effective rate. Littlejohn: 3-layer uniGRU H=512, RNN-T (not CTC) for streaming at

12.5 Hz. N=1 (no cross-patient). Most systems use 12–20 Hz; Metzger's 50 Hz is the outlier (paired with much larger GRU). Supports our configurable downsampling.

Wu 2025 — articulatory features "more robustly encoded in vSMC than acoustic features, learned faster with limited data." P3 was intra-operative (PCC=0.76)—closest analogue to our setting.

2.3 What Doesn't Apply

- **Spike-based FMs** (BIT 367 hrs, NDT3 2000 hrs): Poisson spikes ≠ Gaussian HGA
- **Large-scale SSL**: We have ~3 hrs pooled; minimum successful SSL was ~30 min/pt
- **Dual Pathway speech FM**: Validated for auditory cortex (STG) only

3 Architecture

3.1 Phase 2: Field Standard (E1)

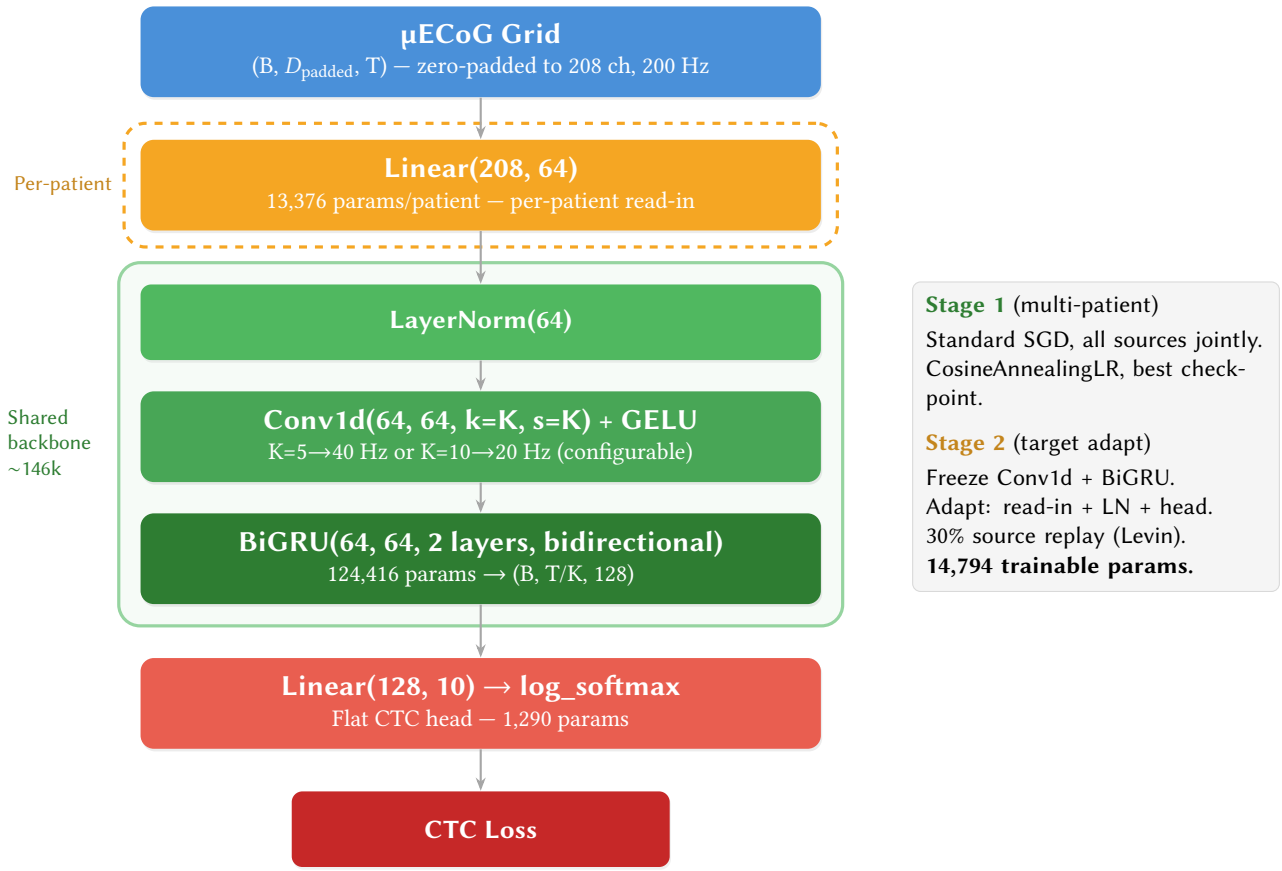

Figure 1: Phase 2 field standard (E1). Training identical to Phase 3.

3.2 Phase 3: Two Innovations (E2)

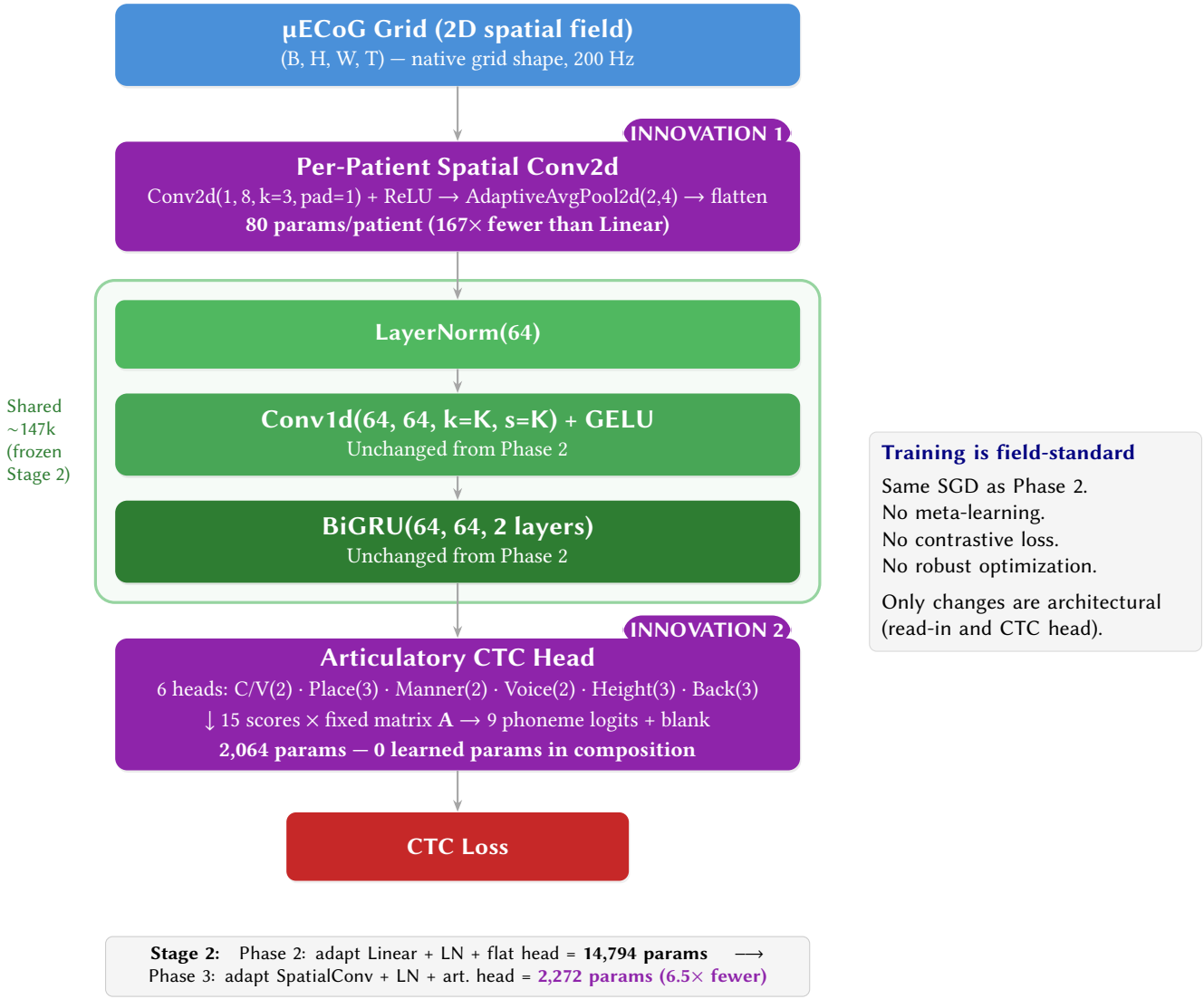

Figure 2: Phase 3 full model (E2). Purple = novel. Green = shared (frozen Stage 2). Training identical to Phase 2.

4 The Two Innovations

4.1 Innovation 1: Per-Patient Spatial Convolution

Insight: uECOG grids are 2D spatial fields—each timepoint is a low-resolution "image" of cortical activity. No one in the field treats them this way; everyone flattens to an unordered vector.

	Phase 2: Linear	Phase 3: Spatial Conv
Per-patient params	13,376	**80** (167× fewer)
Spatial structure	Ignored	Exploited (2D conv)
Grid size handling	Zero-pad to max	AdaptiveAvgPool (native)
Inductive bias	None	Translation equivariance
Stage 2 trainable	14,794	**2,272** (6.5× fewer)

Per-patient because array orientation is *not consistent* across patients—electrode (1,1) maps to different anatomical locations. Both 8×16 and 12×22 grids pool to $(2, 4)$ transparently. Duraivel 2023: HG correlation drops below r=0.6 at <2 mm, confirming adjacent electrodes carry distinct spatial information.

Why this matters more for uECOG than Utah: Boccato's "mostly diagonal" transforms reflect Utah's fixed placement (variation ≈ gain). Our variable surgical placement means cross-patient variation includes spatial remapping—exactly what Conv2d's local spatial kernels capture, with far fewer parameters than a full affine matrix.

4.2 Innovation 2: Articulatory CTC Head

Insight: Motor cortex encodes articulatory gestures, not phoneme categories. Somatotopic organization maps to place of articulation (lips → tongue → larynx). Wu 2025: articulatory features "more robustly encoded in vSMC than acoustic features." Duraivel 2023: confusion errors track Chomsky–Halle distance.

Six parallel sub-heads classify articulatory features:

Head	Classes	Distinguishes
Consonant/Vowel	2	/b,p,v,g,k/ vs. /a,æ,i,u/
Place	3	bilabial vs. labiodental vs. velar
Manner	2	stop vs. fricative
Voicing	2	voiced vs. voiceless
Height	3	low vs. mid vs. high
Backness	3	front vs. central vs. back

15 feature scores compose into 9 phoneme logits via a *fixed* binary matrix **A** (zero learned parameters in composition). Key property: /b/ vs /p/ = voicing score alone—the model *must* learn voicing to distinguish them.

4.3 Component Reference Table

Component	Status	Supporting papers	Reasoning
Per-patient input layers	Standard	Willett, Boccato, Levin, Nason, Singh, BIT	Absorbs cross-patient recording variation. Levin: shared input makes transfer worse
BiGRU 2×64 backbone	Standard	Singh (BiLSTM 2×64, 36/44 ph), Willett/Metzger/Littlejohn (H=500–512, 50–150× data). BIT: transformer (367 hrs)	RNNs for variable-length sequences; H=64 matches data scale. Transformers need far more data
CTC loss	Standard	Willett, Metzger, Boccato, Levin, Nason, Singh (CTC). Littlejohn (RNN-T)	Alignment without frame-level labels. Universal for phoneme-level decoding
Conv1d temporal downsampling	Standard	Spalding (k=10→20 Hz), Willett (~12.5 Hz), Littlejohn (12.5 Hz), Metzger (50 Hz, outlier)	Most systems 12–20 Hz. Configurable K=5 or K=10
Two-stage LOPO	Standard	Singh (Mode 3), Levin (30% replay), Willett, Nason	Freeze backbone, adapt per-pt layers. Singh: freeze-backbone optimal
LayerNorm	Standard	Chen 2024 (InstanceNorm), BIT (LayerNorm)	Per-sample; no batch stats leak. Unfrozen in Stage 2 (~128 params)
Augmentation stack	Standard	Spalding (±100 ms shift), Willett (noise), Boccato (gain dominant)	Each augmentation simulates a specific cross-patient variation source
Best-checkpoint	Standard	Standard practice (no BCI paper uses SWA)	Simple; no unprecedented complexity
Per-patient Conv2d	**Novel**	*Motivated by:* Chen 2024 (3D ResNet > LSTM on grids), Duraivel 2023 (HG r<0.6 at <2 mm)	First 2D conv on uECOG. 80 vs 13,376 params; spatial inductive bias for remapping
Articulatory CTC head	**Novel**	*Motivated by:* Wu 2025 (articulatory > acoustic in vSMC), Duraivel 2023 (errors ∝ artic. distance), Bouchard 2013	9-way → 6 binary/3-way sub-problems matching motor cortex encoding. Fixed composition

Table 1: Pipeline component justification. 8 of 10 components are field-standard with multi-paper support; 2 novel components are independently ablated (E3–E4).

5 What I Evaluated and Rejected

Three training innovations were evaluated and downgraded to extended ablations (E5–E7):

Reptile meta-learning. Per-patient layers already provide transfer (Singh Mode 3). Backbone frozen in Stage 2 → "easy to adapt" applies to params we don't touch. N=7 is thin for meta-learning.

SupCon. CTC + per-patient layers already enforce alignment (shared head decodes all patients). Time-pooled embeddings lose sequential order. Too few positives per class.

DRO. Hard patients (S3: 46 trials, S5: 63 ch) are hard for *practical* reasons, not neural coding differences. Upweighting = overfitting to noise. N=7 running EMA too noisy. Implicit difficulty-aware training already exists (higher loss → larger gradients).

6 Experiments

6.1 Core Matrix (for paper)

Exp	Read-in	Training	CTC head	Tests
E0	PCA+CCA+SVM	—	Flat (SVM)	Baseline (0.31)
E1	Linear(208,64)	Standard SGD	Flat(128→10)	Field standard
E2	**Spatial Conv2d**	**Standard SGD**	**Articulatory**	**Full model**
E3	Linear(208,64)	Standard SGD	Articulatory	−SpatialConv
E4	Spatial Conv2d	Standard SGD	Flat(128→10)	−ArtHead

Table 2: Training is identical across E1–E4. All differences are architectural.

Attribution: E1−E0 = modernization gain. E2−E1 = innovation gain. E2−E3 = spatial conv contribution. E2−E4 = articulatory head contribution.

Development path: E1 → E1a (+SpatialConv) → E2 (+ArtHead). Only proceed if no PER degradation. Quick validation: 1 LOPO fold, 1 seed per step.

Extended ablations (E5–E14): +DRO, +Reptile, +SupCon, diagonal gain, +SWA, +DANN, +TENT, temporal dilation, pooling strategy, Conv3d hybrid.

Evaluation: PER primary (edit distance / 3). Balanced accuracy secondary. LOPO, 3 seeds. Wilcoxon signed-rank (N=8, $p < 0.05$ requires ≥7/8 improve).

7 Data

Pt	Dx	Array	Trials	Sig ch	Frames	Notes
S1	PD	8×16	144	111	28,800	
S2	PD	8×16	148	111	29,600	
S3	Tumor	12×22	**46**	149	**9,200**	Fewest trials
S4	PD	8×16	151	74	30,200	
S5	PD	8×16	151	63	30,200	Fewest sig ch
S6	Tumor	12×22	137	144	27,400	
S7	Tumor	12×22	141	171	28,200	
S8	Tumor	12×22	178	201	35,600	Most trials/ch

- Task: non-word repetition, 52 CVC/VCV tokens, 3 phonemes each (e.g., /abe/)
- CTC targets: 3-phoneme sequences (e.g., [a, b, e])
- 9 phonemes: /a/, /æ/, /i/, /u/, /b/, /p/, /v/, /g/, /k/
- Paired audio: lavalier mic, MFA labels. MNI-152 coords confirmed
- Array orientation: NOT consistent across patients

8 Cross-Task Pooling (Duraivel 2025)

Pseudo-word repetition task with **same 9 phonemes** in CVC/VCV tokens. At least 3 patients overlap:

Nearly doubles training data for overlapping patients. Same IRB, same preprocessing. Future direction (after E1–E4).

Duraivel	Spalding	Addl trials	Sig ch
D-S1	S8	156–208	201/256
D-S2	S5	156–208	63/128
D-S3	S1/S2	156–208	111/128

9 Questions for Zac

9.1 Blocking (need before coding starts)

1. **Where is the data?** Repo references `pt_decoding_data_S62.pkl` and per-patient `.mat` files — DABI, lab server, or elsewhere? Access process?
2. **Patient ID mapping.** Code (S14, S23, S62…) → paper (S1–S8). What's the mapping table?
3. **Grid spatial convention.** Do `hgTrace` ch_x/ch_y = physical grid rows/columns? Is orientation consistent or arbitrary per patient?
4. **Non-significant channels.** Does `hgTrace` preserve the full 2D grid with non-sig channels zeroed, or are non-sig channels excluded? Conv2d needs full grid with non-sig zeroed.
5. **Epoch width.** Data beyond ±500 ms in `.mat` files? (Need ±600 ms for time-shift augmentation.)
6. **Label encoding.** Are `phonSeqLabels` 0-indexed (0–8) or 1-indexed (1–9)? CTC blank must be index 0.
7. **MNI coordinates and audio.** Where stored? What format? Are MFA phoneme labels already generated?

9.2 Important for project scope

8. **Are there more uECOG patients?** Beyond published 8 — others with good data (phoneme or word/nonword tasks)? Full patient list?
9. **Which patients overlap with Duraivel 2025?** We infer D2025-S1=S8 (201/256 tumor), D2025-S2=S5 (63/128 DBS), D2025-S3=S1 or S2 (111/128 DBS). Correct? Can you disambiguate S1 vs S2?
10. **Is pseudo-word HGA data available?** For overlapping patients — same `.mat` format? Would we need to re-window (±1500 ms → ±500 ms)?

9.3 Nice-to-have

11. **Is Spalding S3 = Duraivel 2023 S4?** S3 (46 trials, 149/256 tumor) is our hardest patient. Excluded from Duraivel 2025 due to low trials? Any known bad-channel patterns?
12. **CCA details.** Pairwise-to-universal projection method? (For baseline reproduction.)
13. **TME surrogate.** W=8, p=0.20 had low power. Other W values tried?

10 Timeline

Compute: 4 core × 8 LOPO × 3 seeds × ~30 min ≈ 48 GPU-hours.

Phase 2 is a checkpoint—publishable standalone result if Phase 3 doesn't pan out.

Phase	Duration	Depends on
Phase 0: Setup	1–2 weeks	Data access (Q1–2)
Phase 1: Reproduce baseline	1 week	Working code + data
Phase 2: Field standard (E1)	**2–3 weeks**	Phase 1
Phase 3: Full model (E2)	1 week	Phase 2 validated
Ablations (E3–E4)	1 week	E2 converged (parallel)
Extended (E5–E14)	1–2 weeks	If time permits
Analysis + figures	1–2 weeks	All experiments done

Full details: `implementation_plan.md`, `research_synthesis.md`, `future_directions.md`